

Mini Project in Electrical Circuits and Systems (EC280)

Topic: “Low-Power Analog Signal Conditioning Front-End for IoT Sensor Nodes”

With the rapid growth of the Internet of Things (IoT) and wearable devices, sensors are being deployed everywhere- from health monitoring to environmental tracking. However, raw sensor signals are often very weak and noisy, making direct processing by microcontrollers inefficient and power-hungry. A low-power analog front-end (AFE) provide an elegant solution by conditioning these signals through amplification and filtering before they reach the digital domain. This not only reduces noise and improves accuracy but also minimizes overall power consumption, which is critical for battery-powered and portable systems.

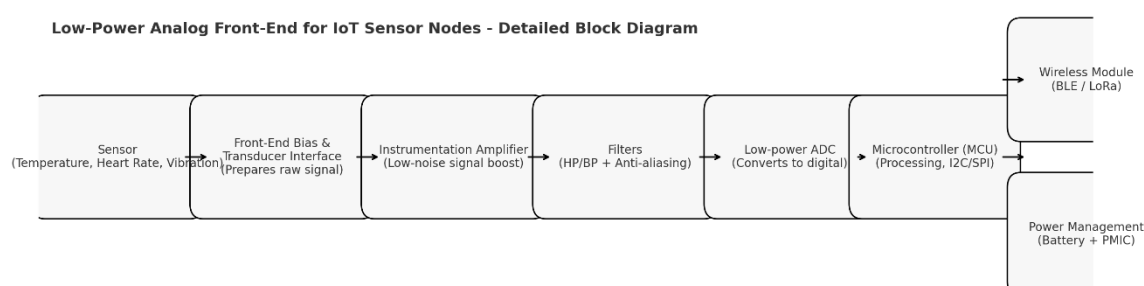
Primary Objective:

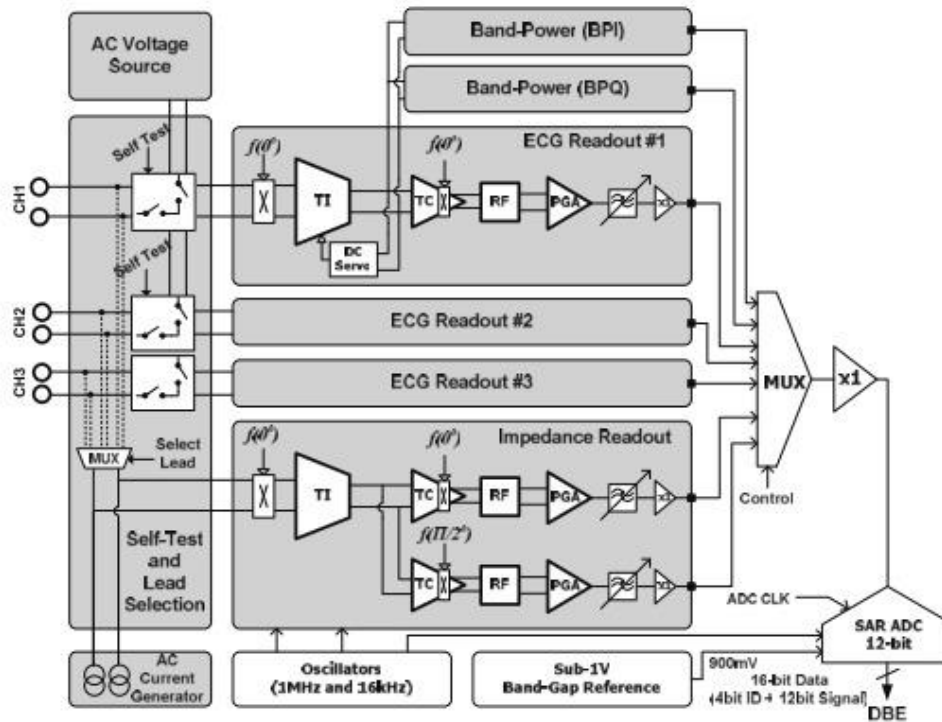
To design and implement an ultra-low-power analog front-end (AFE) for conditioning signals from a sensor (e.g., temperature, heart rate, vibration), with the goal of reducing noise and power consumption before handing off to a microcontroller.

Detailed Goals:

- Implement op-amp-based amplification, filtering (low-pass or band-pass), and possibly instrumentation amplifier stages to preprocess the analog sensor output.
- Optimize for low power—target power consumption in the microwatt range, similar to recent state-of-the-art AFE designs.
- Ensure the filtered analog output is clean and suitable for ADC sampling by an MCU, minimizing noise and interference.
- Enable embedded integration: MCU reads processed signal, displays it or transmits via wireless module.
- Potentially make the design adaptable to different sensing modalities (ECG, environmental sensing, wearables).

Block Diagram of the Complete System





Related Recent IEEE Papers & References

1. **“Design of a Low-Power Ground-Free Analog Front End for ECG Acquisition”** – Kanisorn Watcharapongvinit *et al.*, *IEEE Transactions on Biomedical Circuits and Systems*, Apr. 2023.
Demonstrates an AFE for ECG with only 6.55 μW power and low input-referred noise, using a two-electrode ground-free design in 0.18 μm CMOS.
2. **“A Fully Reconfigurable Universal Sensor Analog Front-End IC for the Internet of Things Era”** – Sung-Hun Jo *et al.*, *IEEE Sensors Journal*, Apr. 2019.
Proposes a reconfigurable AFE that can process R, C, V, and I sensor outputs with a single analog front-end and input bond-pad; uses correlated double sampling (CDS) to reduce low-frequency noise

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